

Sejin Jeon

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Education

Sogang University

2023–present

B.S. candidate in Physics; second major in Mathematics

Seoul, Republic of Korea

- Cumulative GPA: 3.69/4.30; 116 credits completed. Currently fulfilling mandatory military service.
- **Selected coursework:** Real Analysis; Abstract Algebra I–II; Complex Analysis I–II; Topology I; General Relativity; Quantum Field Theory I; Quantum Mechanics I–II.

Canadian International School

2018–2022

International Baccalaureate Diploma Programme

Singapore

Research Interests

Mathematical physics; general relativity; black-hole solutions; gauged supergravity; holography; quantum field theory.

Research Experience

Quantum Gravity and Holography Laboratory

2024–present

Undergraduate Researcher, advised by Prof. Junho Hong

CQeST, Sogang University

- Contributed to two projects on asymptotically AdS black-hole solutions in four- and six-dimensional gauged supergravity.
- For the G-LAMP-funded project *Electrically Charged Rotating Black Holes with Four Independent Charges in the 4D Gauged STU Model*, derived and cross-checked equations of motion, developed Mathematica code to verify known solutions, and contributed toward a four-charge black-hole ansatz.

CERN-Korean Summer Student Programme

June–July 2025

Participant and project presenter

CERN, Geneva, Switzerland

- Completed an intensive programme covering particle physics, quantum field theory, detector systems, experimental data processing, and the CMS and ALICE experiments, with technical visits across CERN facilities.
- Presented *Analysing Particle Decays Through ROOT*: used ALICE open data and ROOT to construct invariant-mass histograms and fit the $K_S^0 \rightarrow \pi^+ \pi^-$ resonance, comparing the extracted decay width with a two-body theoretical estimate.

Honors and Research Funding

G-LAMP Undergraduate Research Grant

April–July 2025

Sogang University G-LAMP Programme, supported by the Ministry of Education. Funded the four-charge rotating black-hole project in gauged STU supergravity.

Honorable Mention (Third Place)

May 2025

2024 Numerical Astrophysics Competition, Korea Astronomy and Space Science Institute (KASI).

Excellent Youth Scholars Award

July 2021

16th Korea Scholars Conference for Youth, Yonsei University. Recognized an astronomy research proposal examining determinations of the Hubble constant and the age of the Universe using the Friedmann equations and high-resolution spectroscopy.

Conferences, Workshops, and Schools

CQeST 2025 Workshop on Cosmology and Quantum Spacetime	August 18–22, 2025
Five-day workshop in Incheon featuring invited, contributed, and CQeST talks on cosmology, modified gravity, string theory, black-hole physics, and astroparticle phenomenology.	
Second IBS CTPU-CGA Winter Camp	February 17–20, 2025
Recommendation-based winter school at IBS in Daejeon, limited to approximately 40 participants, with intensive lectures in English on particle physics, general relativity, and cosmology.	
Third Han-Gang Gravity Workshop	January 16–17, 2025
CQeST workshop at Sogang University featuring lectures on holographic renormalisation and standard black-hole and wormhole solutions.	
Numerical Relativity and Gravitational Waves Winter School	2024 and 2025
Participant, Korea Astronomy and Space Science Institute (KASI).	
Biophysics School	2024
Participant, Asia Pacific Center for Theoretical Physics (APCTP) and POSTECH.	
Workshop for Young Mathematicians in Korea	2024
Participant, Seoul National University.	

Leadership and Service

Byulban Astronomy Club, Sogang University	2025
<i>Head of Student Affairs</i>	<i>Full academic year</i>
Mandatory Military Service, Republic of Korea	Ongoing
<i>Currently fulfilling national service</i>	<i>Republic of Korea</i>

Technical Skills

Scientific computing: Python, MATLAB, Mathematica, ROOT
Research tools: LaTeX, Git and GitHub, Linux, Neovim, Visual Studio Code